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Punctuation Restoration Improves Structure Understanding without Supervision

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Abstract

Unsupervised learning objectives like autoregressive and masked language modeling constitute a significant part in producing pre-trained representations that perform various downstream applications from natural language understanding to conversational tasks. However, despite impressive generative capabilities of recent large language models, their abilities to capture syntactic or semantic structure within text lag behind. We hypothesize that the mismatch between linguistic performance and competence in machines is attributable to insufficient learning of linguistic structure knowledge via currently popular pre-training objectives. Working with English, we show that punctuation restoration as a learning objective improves performance on structure-related tasks like named entity recognition, open information extraction, chunking, and part-of-speech tagging. Punctuation restoration results in $\geq 2\%$ improvement in 16 out of 18 experiments, across 6 out of 7 tasks. Our results show that punctuation restoration is an effective learning objective that can improve structure understanding and yield a more robust structure-aware representations of natural language in base-sized models.

1 Introduction

The modern natural language processing paradigm centers around transformer-based pre-trained language models (PLMs; Peters et al. (2018); Radford et al. (2018); Devlin et al. (2019)). They are optimized on masked language modeling (MLM) and autoregressive language modeling, which provide powerful representations to approach various problems in natural language processing. It is no exaggeration that language models have become effective in tasks like named entity recognition (NER), information extraction, semantic role labeling (SRL) that require understanding of syntactic, semantic, and discourse structure (Wang et al., 2021, 2022). However, the following suggests there

is still room for improvement in current language models' abilities to understand such structure in natural language to perform downstream tasks reliably and robustly.

1. **The reversal or factorization curse.** Language models fail to infer "B is A" from "A is B" (Berglund et al., 2024), or their representations are highly dependent on the order (factorization) of the input (Kitouni et al., 2024).
2. **The curse of performance instability.** Model checkpoint initialization and training dataset order strongly affects sensitivity to syntactic structure (Zhou et al., 2020; McCoy et al., 2020; Du and Nguyen, 2023).
3. **Poor out-of-distribution generalization.** Systems report close-to-human performance on one dataset yet perform poorly on other datasets representing the same task, due to their picking up **spurious correlations** rather than learning the task (Gururangan et al., 2018; McCoy et al., 2019; Serrano et al., 2023).
4. **Insufficient or underutilized structure information.** While PLMs do encode some structure, they are poor few-shot structure predictors (Zhao et al., 2023; Bai et al., 2023) and perform better when input is reinforced with linguistic structure information (Strubell et al., 2018; He et al., 2020; Sachan et al., 2021; Wu et al., 2021; Fei et al., 2021; Xie et al., 2023; Huang et al., 2024). This indicates their representations are insufficient or underutilized.

These four phenomena illustrate that current representations as a result of autoregressive (Radford et al., 2018) or masked (Devlin et al., 2019; Liu et al., 2019; Raffel et al., 2019) language modeling are insufficient for structure understanding.

Efforts to mitigate such shortcomings include data-oriented approaches like syntactic augmentation to improve robustness to spurious correlations (Min et al., 2020; Yaghoobzadeh et al., 2021) and reversing input to mitigate the reversal curse (Golovneva et al., 2024). Architecture oriented efforts include adding explicit graph network layers to encode structure, resulting in improvement in benchmark scores (Zhang et al., 2019; Sachan et al., 2021; Wu et al., 2021) and generalization abilities (He et al., 2020; Sartran et al., 2022).

They are human-in-the-loop methods that require human input or annotation, or a system that requires such annotation. Recent work in distilling linguistic structure knowledge from natural language text to representations without supervision include inside-outside dynamic programming for tree induction (DIORA; Drozdov et al., 2019), dependency-constrained self-attention (Shen et al., 2021; Momen et al., 2023), and augmenting MLM with sentence-level contrastive learning (CLEAR; Wu et al., 2020). With the exception of CLEAR, these methods require additions to the model architecture. Wang et al. (2021) and Wang et al. (2022) propose structure pre-training but use human-annotated data.

In this paper, we investigate whether it is possible for an unsupervised method to mitigate the four shortcomings of the modern language model without implementing additional parser, tree, or graph architecture. In particular, we believe the pre-training stage of current PLMs may be further improved and propose punctuation restoration (PR) as an unsupervised learning objective that improves structure understanding. Punctuation markers, along with capitalization, often serve as boundary markers between different syntactic components of the sentence (Briscoe, 1996; Bayraktar et al., 1998). Punctuation marks also correspond to prosodic features of the sentence (Chafe, 1988), which in turn contain linguistic information (Wilson and Wharton, 2006; Wolf et al., 2023). Thus, the model’s ability to predict punctuation from plain text may correlate to its ability to encode syntactic boundaries and thus structure. We hypothesize that additional optimization on punctuation restoration yields representations with increased sensitivity to structure, measured by in-distribution test set score, out-of-distribution generalization performance, and stability across initialization in structure-related natural language pro-

cessing (NLP) tasks.

Intuitively, punctuation restoration may seem like an easy task. However, predicting punctuation and capitalization given text still remains nontrivial (Păiş and Tufiş, 2022; de Lima et al., 2024; Pang et al., 2024), and is an area of active work especially for post-processing results from automatic speech recognition and transcription systems (Alam et al., 2020; Zhu et al., 2024; You and Li, 2024; Zhong and Sun, 2025). We provide performance on the objective task in Appendix C.1. Although language modeling already includes predicting punctuation and capitalization, explicit optimization on punctuation restoration would allow models to predict them without explicit local context (e.g. beginning of sentence or quotation, following capitalization).

2 Objective and experimental setup

2.1 Objective design

The punctuation restoration objective predicts the original text from its "cleared-formatting" counterpart. In our implementation, we remove the following set of punctuation marks: the comma ,, the period ., the exclamation point !, the question mark ?, the single-quotation mark ', and the double-quotation mark ", along with capitalization, as shown below. Boldface indicates an addition to or a modification of source text.

- Source: lee faker sang-hyeok (hangul: 이상혁) is a league of legends esports player currently mid laner and part owner at t1
- Target: Lee **“Faker”** Sang-hyeok (Hangul: 이상혁) is a **L**eague of **L**egends esports player, currently mid laner and part owner at **T**1.

While it is possible that a different selection yield better results, our selection reflects frequency (Sun and Wang, 2019) as well as syntactic significance (Bayraktar et al., 1998; Brabanter, 2023).

Similarly to popular pre-training objectives like MLM, autoregressive language modeling, and next-sentence prediction, the objective requires no human input. The objective is also architecture-agnostic and can be easily modified as appropriate.

From an internal database of English news articles, accessed between January 2022 and August 2023, we collected a total of 437,031 article excerpts, which are non-overlapping parts separated by a limiting word count of 150. One thousand excerpts each are used as the development and test

sets, while the remaining 435,031 excerpts are used for training.

2.2 Experimental setup

Our experiments involve two stages. In the first stage, we take the pre-trained weights of the T5-base¹ model (Raffel et al., 2019), and perform additional pre-training on the punctuation restoration objective to produce PR-T5. Then, in the second stage, we fine-tune PR-T5 on downstream tasks and datasets.

In the first stage, the model f is given the "cleared-formatting" token sequence x comprising of tokens x_t and optimized to predict the original, fully punctuated and capitalized text y comprising of tokens y_t as described in Section 2.1. However, since there is textual overlap between x and y , assuming trivial copy error rate, we can write the model f as a predictor of capitalization and punctuation information $m_t = y_t - x_t$:

$$m_t = f(x, y_{<t}) = \begin{cases} \phi \\ \text{addPunct}(x_t, \theta) \\ \text{addCap}(x_t, \theta) \end{cases}$$

Thus, the effective loss is as follows:

$$\mathcal{L} \approx -\frac{1}{N} \sum_{t=1}^N \log P(m_t | x, y_{<t}).$$

In the second stage, we fine-tune PR-T5 and measure the effects of punctuation restoration in downstream tasks. We measure effects across 13 datasets that represent 7 tasks² and across 3 settings: generative, discriminative, and multi-task. In the generative setting, fine-tuned PR-T5 makes entity or tag predictions via autoregressive generation. We conduct 16 experiments in the generative setting, with 13 datasets from 7 tasks. In the multitask setting, fine-tuned PR-T5 is trained to make predictions for two tasks at once, namely NER and Open Information Extraction (OpenIE). We conduct 1 experiment in the multitask setting, with 2 datasets from 2 tasks. Generative and multitask predictions are illustrated in Table 4. In the discriminative setting, PR-T5’s decoder block is replaced with a classification head, as described in Appendix A.1 and Figure 1. We conduct 1 experiment in the discriminative setting, with 1 dataset from 1 task. We fine-tune the publicly available pre-trained T5 weights on

the same downstream tasks and use their performance as comparison baseline for all three settings. We publicly release our [architecture](#), [training](#), and [inference code](#).

3 Results

We measure the effects of punctuation restoration as an additional pre-training objective on downstream tasks on t5-base, with the four behaviors outlined in Section 1 in mind. In this section, we find direct evidence that this method helps mitigate three out of four behaviors we describe in Section 1.

We report our results in Tables 1, 2, 3. Each reported value of precision, recall, and F1 represents an average over the same 5 seed initializations, with the exception of discriminative NER, where we analyze 15 seed initializations.

3.1 Structure information encoding and use

In all 18 experiments across dataset, task, and setting, PR-T5 reports improved performance over T5 baselines. Among them, 16 experiments report improvements $\Delta \geq .02$, and 10 experiments $\Delta \geq .05$ (Tables 1, 2, 3). This is evidence that punctuation restoration makes available a nontrivial amount of structure information that previously may have been unavailable or underutilized, mitigating behavior 4 from Section 1.

3.2 Performance stability and out-of-distribution generalization

An out-of-distribution evaluation measures performance on a dataset that represents the same task but comes from a different source than the training dataset (e.g. evaluating on CaRB (Bhardwaj et al., 2019) after fine-tuning on OIE2016 (Stanovsky and Dagan, 2016)). It is an effective measure of robustness of a representation, as fine-tuned models often learn the dataset, rather than learning the task (Gururangan et al., 2018; McCoy et al., 2019; Serrano et al., 2023). We compare out-of-distribution generalization ability of PR-T5 to that of T5 in 5 experiments across NER, OpenIE, Chunking, and POS tagging, where we observe $\Delta \geq .05$ increase in 4 of them (Table 1). This is evidence that punctuation restoration improves out-of-distribution generalization, mitigating behavior 3 in Section 1.

In addition, we observe that punctuation restoration reduces performance instability. Compared to T5, PR-T5’s distribution of NER performance

¹See Appendix C.1 for details on model size selection

²See Appendix B for task and dataset details

Task	Training set	Evaluation set	t5-base			+ PR			Δ
			P	R	F1	P	R	F1	F1
NER	Econ-mNER	ID	.69	.65	.67	.90	.89	.89	▲.22
		Econ-sNER	.67	.76	.71	.74	.81	.77	▲.06
	GENIA	ID	.57	.73	.64	.64	.76	.69	▲.05
	CoNLL03	ID	.89	.90	.89	.92	.92	.92	▲.03
OpenIE	EconIE-PRO	ontonotes	.87	.88	.88	.91	.91	.91	▲.03
		ID	.47	.43	.45	.60	.63	.62	▲.17
	OIE2016	CaRB	.22	.16	.19	.62	.42	.50	▲.31
		ID	.16	.19	.18	.19	.19	.19	●.01
Chunking	CoNLL00	CaRB	.10	.15	.12	.26	.27	.27	▲.15
		ID	.94	.94	.94	.96	.96	.96	▲.02
SRL	CoNLL12	CoNLL03	.41	.41	.41	.41	.42	.42	●.01
		ID	.75	.79	.77	.84	.86	.85	▲.08
SBD	PTB	ID	.97	.72	.81	.98	.98	.98	▲.17
POS	CoNLL00	ID	.96	.96	.96	.98	.98	.98	▲.02
		CoNLL03	.74	.87	.79	.84	.88	.86	▲.07
RE	TACRED	ID			.67			.83	▲.16

Table 1: Our main results where we compare t5-base model to PR-t5-base (+PR). ID denotes in-distribution evaluation on a dataset from the same source as the training set. See Appendix B for dataset details.

	t5-base (joint)			+ PR			Δ
	P	R	F1	P	R	F1	F1
NER	.86	.84	.85	.87	.86	.87	▲.02
OIE	.57	.60	.58	.60	.62	.61	▲.03

Table 2: Multitask (Econ-mNER, EconIE-PRO) performance.

	t5-base (EO)			+ PR			Δ
	P	R	F1	P	R	F1	F1
min	.67	.91	.78	.74	.90	.82	▲.04
max	.88	.94	.91	.90	.94	.91	●.00
avg	.78	.93	.85	.83	.92	.88	▲.03
sdev	.061	.009	.035	.048	.010	.027	▼.008

Table 3: Discriminative Econ-mNER performance.

across initialization seeds is narrower. Minimum-maximum range (▼.04) and standard deviation (▼23%) both decrease with additional pre-training in punctuation restoration, as reported in Table 3. The results support our hypothesis that punctuation restoration increases stability across initialization seed and training dataset order, mitigating behavior 2 discussed in Section 1.

4 Discussion

Results from Section 3 support our hypothesis that complementing MLM with a more structure-related objective improves structure understanding. In par-

ticular, we use a punctuation restoration objective, described in Section 2 and evaluate with various structure-related tasks. While it is difficult to investigate the exact mechanism of how additional training on punctuation restoration improves learned representations, we attempt to provide an explanation.

In Section 1, we analyze that current methods for representation learning during the pre-training stage lack sufficient signal, and hypothesize additional training with a structure-sensitive objective should improve structure understanding. Much like how prosody helps disambiguate syntax in human speech processing (Price et al., 1991; Kahn et al., 2005), punctuation can be a useful guide in syntax disambiguation, and eventually toward forming a robust representation of text. Punctuation marks represent prosodic design (Chafe, 1988) which carries linguistic information (Wilson and Wharton, 2006; Wolf et al., 2023). Punctuation marks often also indicate syntactic or semantic boundaries (Briscoe, 1996; Bayraktar et al., 1998). Optimizing a computational system to predict punctuation allows it to predict syntactic and semantic boundaries, even in the absence of punctuation in the original text. Sufficient training in restoring punctuation can imitate effects of explicitly providing a parse, facilitating natural language understanding via a stronger understanding of sentence structure.

Performance improvement from punctuation

restoration is not limited to a specific dataset, task, and setting³, and represents an overall increase in representation robustness, as we observe out-of-distribution performance jump in NER, OpenIE, and chunking. Because of the wide range of experiments in which improvement is observed, we interpret this to be a general improvement of structure understanding rather than fortunate task-specific artifacts from the additional training.

Our methods yield a more reliable and robust representation that can be easily implemented and do not interfere with architectural additions. Punctuation restoration can be applied to reinforce structure understanding and improve robustness of learned representations regardless of model choice, or task-specific engineering policy. The effective objective requires no supervision, and one can construct a training corpus with little computational or manual resources.

Limitations

The idea of structure understanding reinforcement via punctuation restoration is still young—many decisions relevant to the learning objective in this paper, including selection of punctuation marks and source of learning corpus warrant additional investigation in future work. Our set of training hyper-parameters also will benefit from additional attention.

Among the 4 behaviors discussed in Section 1, we find direct evidence that punctuation restoration mitigates only three of them. While we predict that unsupervised structure learning via objectives like punctuation restoration can help mitigate the reversal (factorization) curse, this will need explicit verification.

While our experiments show promise in base-sized natural language understanding models for English, its effects in larger models, implications to generative or conversational systems, and generalization to other languages and thus language-agnostic nature also need to be verified.

It is also likely that punctuation restoration is not the only unsupervised learning objective that can be used to improve the representation learning stage of training NLP systems. Other forms of unsupervised structure learning, possibly simpler and more effective methods than punctuation restoration, as well as optimizations on objective

combination (e.g. with word prediction methods) should be studied in future work.

Finally, although we believe that additional optimization on punctuation restoration improves the models’ encoded linguistic structure, leading to performance jumps between T5 and PR-T5, we do not control for the additional compute or the exposure to novel data in this paper. Future work explicitly controlling for such variables will provide more robust arguments for punctuation restoration as a representation learning objective.

Responsible research statement

We use OpenAI’s GPT-3.5 Turbo (Brown et al., 2020) as a punctuation restoration performance baseline, and as a debugging assistant during the project’s technical implementation.

The Econ-mNER dataset was annotated by paid, full-time employees who are trained linguists knowledgeable about their work and the dataset’s downstream use. They are compensated similarly to the region’s 2021 median income level. Their work has been reviewed by an internal board to not contain any personally identifiable information. Other internal datasets did not require manual annotation.

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³And decoding method, discussed in Appendix B

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A Additional details on experimental setup

We train the model on the punctuation restoration objective for 40 epochs, before fine-tuning with supervised datasets for downstream tasks. The experiments are run on a single V100 GPU with 32GB VRAM, with half precision and gradient accumulation enabled at 16. Our choice of hyper-parameters are as follows: batch size 32, maximum sequence length 256, learning rate $3e-4$, maximum grad norm 0.5, and Adam epsilon $1e-8$. Number of fine-tuning epochs was 10, with the exception of SRL, which is fine-tuned for 1 epoch only. The additional pre-training lasts about 2 weeks, while the length of each epoch of training varies across datasets between 10 minutes and around 2 hours.

A.1 Discriminative approach

While there exist sophisticated attempts to incorporate the decoder layers in producing a discriminative model from a pre-trained encoder-decoder architecture (Liu et al., 2022), we use a simple architecture where we forgo the decoder block and place a T5ClassificationHead on top of the encoder block of the T5 model. That is, we take the hidden state output from model’s encoder and use it as input to the classification head. An illustration of the model architecture is shown in Figure 1. After additional pre-training on punctuation restoration objective, the decoder block of the t5-base model is removed and a newly initialized classification head is placed on top of the encoder block. The architecture is comparable to those of BERT-like encoder-only models. Even by retaining weights from the encoder blocks only, we observe that additional unsupervised structure learning via punctuation restoration results in downstream task performance improvement.

A.2 Joint multitask generative approach

The joint multitask approach, where we focus on open information extraction using the EconIE-PRO dataset and NER using the Econ-mNER dataset, is similar to the generative approach. The input sequence is identical to the experiments from Section 3, but the output sequence is a concatenation of output sequences from the two datasets, as illustrated in Table 4.

B Additional details on dataset

We use a suite of structure-related NLP tasks to measure model structure understanding. Relevant tasks include named entity recognition (NER), sentence boundary detection (SBD), open information extraction (OpenIE), chunking, semantic role labeling (SRL), part-of-speech tagging, and relation classification. Our selection mostly follows that from Wang et al. (2021) and Lee et al. (2024). We use both public and internal datasets, and check for in- and out-of-distribution generalization. A full list of datasets for each task is shown in Table 5. In the main body of the paper, we discuss effects of punctuation restoration across task, dataset, and setting. Here, we discuss another variable across which punctuation restoration is effective: decoding method.

B.1 Entity generation tasks

NER, OpenIE, SRL, and relation classification are entity generation tasks, where fine-tuned models autoregressively generate entity objects. For example, (Faker: PER), (Faker, is, a League of Legends esports player), (Faker, employeeAt, T1) are NER, OpenIE, and relation classification examples, respectively. The order in which entities are generated does not affect evaluation in the case of entity generation tasks.

Source	Faker is a League of Legends esports player, currently mid laner and part owner at T1.
OpenIE	(Faker, is, a League of Legends esports player) (Faker, is mid laner and part owner at, T1)
NER	(Faker: PER) (T1: ORG)
Multitask	(Faker: PER) (Faker, is, a League of Legends esports player) (Faker, is mid laner and part owner at, T1) (T1: ORG)

Table 4: Example output from generative NER, OpenIE, and multitask models.

B.2 Tag sequence generation tasks

Chunking and POS tagging are tag sequence generation tasks, where fine-tuned models autoregressively generate tag sequences. "NP VP ADVP PP NP NP NP" and "NP VBZ DT NP IN NP" are example sequences of chunking and POS tagging, respectively.

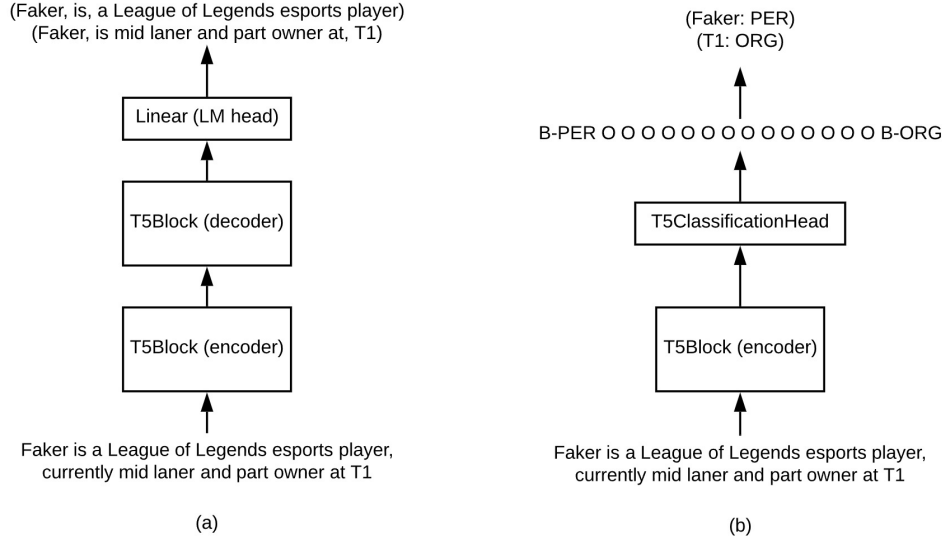


Figure 1: (a) The t5 architecture for a generative, text-to-text approach to NLP tasks. Here, we illustrate open information extraction. (b) A modification to the t5 architecture to allow a discriminative approach to NLP tasks. Here, we illustrate named entity recognition.

B.3 Sequence generation tasks

Punctuation restoration and sentence boundary detection are sequence generation tasks. Fine-tuned models auto-regressively generate natural text sequences, with predefined tags to perform the task. For example, a sentence boundary detection model would generate a [`<s>`] token between sentences, given a passage.

B.4 Token classification tasks

NER in the discriminative setting is a token classification task. Given a sentence of length n , the fine-tuned model outputs an array of length n , each element of which represents whether its corresponding token is part of a named entity. For example, one from a tag set such as [`O`, `B-PER`, `I-PER`, `B-LOC`, `I-LOC`, `B-ORG`, `I-ORG`], as illustrated in Figure 1.

C Additional details on results

In our results, improvements from punctuation restoration persist across decoding methods—entity generation in NER, OpenIE, SRL, and relation classification; tag sequence generation in chunking and POS tagging; sequence generation in sentence boundary detection; and token classification in discriminative NER.

C.1 Objective results

Punctuation restoration, along with capitalization restoration is no trivial task, especially when the model needs to predict restoration location without local context (Gravano et al., 2009; Alam et al., 2020; Păiș and Tufiş, 2022; de Lima et al., 2024; Zhong and Sun, 2025). Should our hypothesis hold, it is likely that syntactic signals from punctuation restoration transfer more effectively in models with stronger punctuation restoration performances. We experiment with three sizes of the T5 architecture (Raffel et al., 2019). We consider t5-small, t5-base, and t5-large. Table 6 includes their punctuation restoration performance, in addition to ChatGPT’s (Brown et al., 2020) zero-shot performance as a reference point, which shows that the objective is nontrivial.

Across the T5 models, there is some correlation between size and punctuation restoration performance. Because the performance gap between t5-base and t5-large models is small ($\Delta 0.00$), while gap between t5-small and t5-base more significant ($\Delta 0.05$), we use the t5-base model for our experiments.

We also note that our selection of the T5 model is due to its ability to perform both generative and discriminative tasks after single pre-training.

Task	Dataset	Source	Task type
Internal datasets			
PR	finPR	Rule-based tagging on finance news	Seq. gen.
NER	Econ-mNER	Manual tagging on finance news and corporate filings	Ent. gen., Tok. cls.
	Econ-sNER	Semi-supervised tagging on finance news	Ent. gen.
OpenIE	EconIE-PRO	Rule-based tagging on finance news, predicate range optimized	Ent. gen.
Public datasets			
NER	GENIA	Kim et al. (2003)	Ent. gen.
	CoNLL 2003	Tjong Kim Sang and De Meulder (2003)	Ent. gen.
	ontonotes	Weischedel et al. (2013)	Ent. gen.
SBD	PTB	Marcus et al. (1993)	Seq. gen.
OpenIE	OIE2016	Stanovsky and Dagan (2016)	Ent. gen.
	CaRB	Bhardwaj et al. (2019)	Ent. gen.
Chunk, POS	CoNLL 2000	Tjong Kim Sang and Buchholz (2000)	Tag gen.
	CoNLL 2003	Tjong Kim Sang and De Meulder (2003)	Tag gen.
SRL	CoNLL 2012	Pradhan et al. (2012)	Ent. gen.
ORE	TACRED	Zhang et al. (2017)	Ent. gen.

Table 5: We use a total of 14 datasets across 8 tasks, including punctuation restoration. Four are internal datasets, while the rest are publicly available.

Model architecture	P	R	F1
ChatGPT 0-shot*	.75	.71	.73
t5-small	.91	.86	.88
t5-base	.93	.92	.93
t5-large	.94	.93	.93

Table 6: Punctuation restoration performance after 50 epochs (small), 40 epochs (base), and 20 epochs (large) of training respectively. *Measured on a small subset of the punctuation restoration evaluation dataset.

punctuation restoration non-trivially transfers to the discriminative approach as well, where the decoder block is removed from the model, as illustrated in Figure 1. Although the maximum performance for T5 and PR-T5 are similar at .91 (●.00), there is a significant difference in the minimum, at .78 and .82, respectively (▲.04). Punctuation restoration results in not only higher performance, but also more consistent and stable sets across different initializations.

C.2 Joint multitask generative setting

Similarly to the generative approach, we observe that additional unsupervised structure learning via punctuation restoration results in downstream task performance improvement (▲.02 NER and ▲.03 OpenIE). While PR-T5 multi-task performance slightly degrades compared to its single-task generative setting (▼.02 NER and ●.01 OpenIE), multitask-T5 outperforms single task-T5 on EconIE-PRO, an open information extraction dataset (▲.13).

C.3 Discriminative setting

Given the results from the single-task generative approach, the transfer from punctuation restoration to multi-task generative approach may be no big surprise, as there is no drastic difference between the generative nature of the two approaches. However, we report that our improved representations from